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-- =====
-- MySQL Joins Practice Assignment
-- Database : d16
-- Topics  : INNER JOIN, LEFT JOIN, RIGHT JOIN, SELF JOIN
-- =====
```

```
-- — Step 0 : Select the working database —————
USE d16;
```

```
-- =====
-- DATASET SETUP
-- =====
```

```
-- — Table 1 : departments —————
CREATE TABLE departments (
    dept_id INT PRIMARY KEY,
    dept_name VARCHAR(30)
);
```

```
INSERT INTO departments VALUES
(1, 'IT'),
(2, 'HR'),
(3, 'Finance'),
(4, 'Sales');
```

```
-- — Table 2 : employees —————
CREATE TABLE employees (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    dept_id INT,
    salary DECIMAL(10,2)
);
```

```
INSERT INTO employees VALUES
(101, 'Ravi', 1, 50000),
(102, 'Priya', 2, 45000),
(103, 'Arjun', 1, 60000),
(104, 'Sneha', 4, 55000),
(105, 'Kiran', NULL, 65000);
```

```
-- — Table 3 : projects —————
CREATE TABLE projects (
    project_id INT PRIMARY KEY,
    project_name VARCHAR(50),
    dept_id INT
);
```

```
INSERT INTO projects VALUES
(1, 'CRM System', 1),
(2, 'Recruitment Portal', 2),
(3, 'Payroll System', 3),
(4, 'Sales Dashboard', 4),
(5, 'Inventory App', NULL);
```

```
-- — Verify data —————
SELECT * FROM departments;
SELECT * FROM employees;
SELECT * FROM projects;
```

```

-- =====
-- TASK 1 : Employee names with their department names
--      using INNER JOIN
-- =====
-- INNER JOIN returns only rows where dept_id matches in BOTH tables.
-- Kiran (dept_id = NULL) will NOT appear because NULL has no match.

SELECT
    e.emp_name,
    d.dept_name
FROM employees e
INNER JOIN departments d
    ON e.dept_id = d.dept_id;

-- =====
-- TASK 2 : Employee ID, employee name, and department name
--      using INNER JOIN
-- =====

SELECT
    e.emp_id,
    e.emp_name,
    d.dept_name
FROM employees e
INNER JOIN departments d
    ON e.dept_id = d.dept_id;

-- =====
-- TASK 3 : All employees INCLUDING those not assigned to any
--      department → LEFT JOIN
-- =====
-- LEFT JOIN keeps ALL rows from the LEFT table (employees).
-- Kiran's dept_name will show as NULL.

SELECT
    e.emp_id,
    e.emp_name,
    d.dept_name
FROM employees e
LEFT JOIN departments d
    ON e.dept_id = d.dept_id;

-- =====
-- TASK 4 : All departments EVEN IF no employees belong to them
--      → RIGHT JOIN
-- =====
-- RIGHT JOIN keeps ALL rows from the RIGHT table (departments).
-- 'Finance' has no employees → emp columns will show NULL.

SELECT
    e.emp_id,
    e.emp_name,
    d.dept_name
FROM employees e
RIGHT JOIN departments d
    ON e.dept_id = d.dept_id;

-- =====
-- TASK 5 : Employees NOT assigned to any department

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-- =====
-- Use LEFT JOIN and filter where the dept side is NULL.
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```
SELECT
    e.emp_id,
    e.emp_name
FROM employees e
LEFT JOIN departments d
    ON e.dept_id = d.dept_id
WHERE d.dept_id IS NULL;
```

```
-- =====
-- TASK 6 : Department names and project names
--         using INNER JOIN
-- =====
```

```
SELECT
    d.dept_name,
    p.project_name
FROM departments d
INNER JOIN projects p
    ON d.dept_id = p.dept_id;
```

```
-- =====
-- TASK 7 : All projects INCLUDING those not assigned to any
--           department → LEFT JOIN
-- =====
-- 'Inventory App' (dept_id = NULL) will appear with dept_name as NULL.
```

```
SELECT
    p.project_id,
    p.project_name,
    d.dept_name
FROM projects p
LEFT JOIN departments d
    ON p.dept_id = d.dept_id;
```

```
-- =====
-- TASK 8 : Employee names, department names, and project names
--           using multiple INNER JOINS
-- =====
-- employees → departments → projects (all linked via dept_id)
```

```
SELECT
    e.emp_name,
    d.dept_name,
    p.project_name
FROM employees e
INNER JOIN departments d
    ON e.dept_id = d.dept_id
INNER JOIN projects p
    ON d.dept_id = p.dept_id;
```

```
-- =====
-- TASK 9 : Departments that currently have NO employees
-- =====
-- Use RIGHT JOIN (all depts) and filter where employee side is NULL.
```

```
SELECT
```

```

d.dept_id,
d.dept_name
FROM employees e
RIGHT JOIN departments d
  ON e.dept_id = d.dept_id
WHERE e.emp_id IS NULL;

```

```

-- =====
-- TASK 10 : Each employee along with their manager's name
--      using SELF JOIN
-- =====
-- A SELF JOIN joins a table to itself.
-- Here employees table acts as both employee AND manager.
-- We add a manager_id column first, then join on it.

-- Step A : Add manager_id column to employees
ALTER TABLE employees ADD COLUMN manager_id INT;

-- Step B : Assign managers
-- Ravi (101) → manager: Arjun (103)
-- Priya (102) → manager: Arjun (103)
-- Arjun (103) → no manager (he is the senior)
-- Sneha (104) → manager: Arjun (103)
-- Kiran (105) → manager: Arjun (103)
UPDATE employees SET manager_id = 103 WHERE emp_id IN (101, 102, 104, 105);
UPDATE employees SET manager_id = NULL WHERE emp_id = 103;

-- Step C : Self Join — alias the table twice
SELECT
  e.emp_name AS employee_name,
  m.emp_name AS manager_name
FROM employees e
LEFT JOIN employees m
  ON e.manager_id = m.emp_id;

```

Screen shots

```

Query OK, 5 rows affected (0.0408 sec)

Records: 5  Duplicates: 0  Warnings: 0
MySQL localhost:33060+ ssl d16 SQL > SELECT * FROM departments;
+-----+-----+
| dept_id | dept_name |
+-----+-----+
| 1 | IT |
| 2 | HR |
| 3 | Finance |
| 4 | Sales |
+-----+-----+
4 rows in set (0.0010 sec)

MySQL localhost:33060+ ssl d16 SQL > SELECT * FROM employees;
+-----+-----+-----+-----+
| emp_id | emp_name | dept_id | salary |
+-----+-----+-----+-----+
| 101 | Ravi | 1 | 50000.00 |
| 102 | Priya | 2 | 45000.00 |
| 103 | Arjun | 1 | 60000.00 |
| 104 | Sneha | 4 | 55000.00 |
| 105 | Kiran | NULL | 65000.00 |
+-----+-----+-----+-----+
5 rows in set (0.0016 sec)

MySQL localhost:33060+ ssl d16 SQL > SELECT * FROM projects;
+-----+-----+-----+
| project_id | project_name | dept_id |
+-----+-----+-----+
| 1 | CRM System | 1 |
| 2 | Recruitment Portal | 2 |
| 3 | Payroll System | 3 |
| 4 | Sales Dashboard | 4 |
| 5 | Inventory App | NULL |
+-----+-----+-----+
5 rows in set (0.0017 sec)

```

```
MySQL localhost:33060+ ssl d16 SQL> SELECT
-> e.emp_name,
-> d.dept_name
-> FROM employees e
-> INNER JOIN departments d
-> ON e.dept_id = d.dept_id;
```

emp_name	dept_name
Ravi	IT
Priya	HR
Arjun	IT
Sneha	Sales

4 rows in set (0.0012 sec)

```
MySQL localhost:33060+ ssl d16 SQL> SELECT
-> e.emp_id,
-> e.emp_name,
-> d.dept_name
-> FROM employees e
-> INNER JOIN departments d
-> ON e.dept_id = d.dept_id;
```

emp_id	emp_name	dept_name
101	Ravi	IT
102	Priya	HR
103	Arjun	IT
104	Sneha	Sales

4 rows in set (0.0015 sec)

```
MySQL localhost:33060+ ssl d16 SQL> SELECT
-> e.emp_id,
-> e.emp_name,
-> d.dept_name
-> FROM employees e
-> LEFT JOIN departments d
-> ON e.dept_id = d.dept_id;
```

1 row in set (0.0011 sec)

```
MySQL localhost:33060+ ssl d16 SQL> SELECT
-> d.dept_name,
-> p.project_name
-> FROM departments d
-> INNER JOIN projects p
-> ON d.dept_id = p.dept_id;
```

dept_name	project_name
IT	CRM System
HR	Recruitment Portal
Finance	Payroll System
Sales	Sales Dashboard

4 rows in set (0.0016 sec)

```
MySQL localhost:33060+ ssl d16 SQL> SELECT
-> p.project_id,
-> p.project_name,
-> d.dept_name
-> FROM projects p
-> LEFT JOIN departments d
-> ON p.dept_id = d.dept_id;
```

project_id	project_name	dept_name
1	CRM System	IT
2	Recruitment Portal	HR
3	Payroll System	Finance
4	Sales Dashboard	Sales
5	Inventory App	NULL

5 rows in set (0.0018 sec)

```
MySQL localhost:33060+ ssl d16 SQL> SELECT
-> e.emp_name,
-> d.dept_name,
-> p.project_name
-> FROM employees e
-> INNER JOIN departments d
-> ON e.dept_id = d.dept_id
-> INNER JOIN projects p
-> ON d.dept_id = p.dept_id;
```

project_id	project_name	dept_name
1	CRM System	IT
2	Recruitment Portal	HR
3	Payroll System	Finance
4	Sales Dashboard	Sales
5	Inventory App	NULL

5 rows in set (0.0018 sec)

```
MySQL localhost:33060+ ssl d16 SQL> SELECT
-> e.emp_name,
-> d.dept_name,
-> p.project_name
-> FROM employees e
-> INNER JOIN departments d
-> ON e.dept_id = d.dept_id
-> INNER JOIN projects p
-> ON d.dept_id = p.dept_id;
```

emp_name	dept_name	project_name
Arjun	IT	CRM System
Ravi	IT	CRM System
Priya	HR	Recruitment Portal
Sneha	Sales	Sales Dashboard

4 rows in set (0.0023 sec)

```
MySQL localhost:33060+ ssl d16 SQL> SELECT
-> d.dept_id,
-> d.dept_name
-> FROM employees e
-> RIGHT JOIN departments d
-> ON e.dept_id = d.dept_id
-> WHERE e.emp_id IS NULL;
```

dept_id	dept_name
3	Finance